

Omar Alaa

AI Engineer | Machine Learning Engineer | NLP Engineer

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PROFESSIONAL SUMMARY

Artificial Intelligence student expected to graduate in 2027 with hands-on experience in Machine Learning, Natural Language Processing, Computer Vision, and DevOps. Experienced in building AI-powered applications using Python, PyTorch, Flask, and Docker, with practical knowledge of CI/CD automation using Jenkins. Passionate about Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and developing scalable, production-ready AI systems.

EDUCATION

Badr University in Assiut (BUA)

Bachelor's Degree in Artificial Intelligence — Faculty of Artificial Intelligence and Data Science

- Relevant Coursework: Machine Learning, Deep Learning, Digital Image Processing, Computer Vision, Embedded Systems, Computer Architecture, Data Structures, Algorithms

EXPERIENCE

Python & DevOps Intern

May 2026

- Implemented CI/CD pipelines using Jenkins to automate software deployment
- Developed Python automation scripts to streamline operational and deployment tasks
- Built and managed Docker containers for consistent application deployment
- Configured and maintained Ubuntu Linux servers for containerized applications
- Collaborated with engineers to troubleshoot deployment issues and improve DevOps workflows

Natural Language Processing (NLP) Trainee

Completed July 2026

National Telecommunication Institute (NTI)

- Completed intensive training in Natural Language Processing and Large Language Models
- Developed an AI-powered Resume Analyzer using Python, Streamlit, SentenceTransformers, and Google Gemini
- Applied semantic similarity techniques to compare resumes with job descriptions
- Worked with embeddings, prompt engineering, and LLM-based resume evaluation

CERTIFICATIONS

- Python & DevOps Development (80 Hours) – Ideospace, in partnership with Assiut Oil Refining Company & ITIDA
- Natural Language Processing (NLP) Training – National Telecommunication Institute (NTI), July 2026
- AI Fundamentals Training – Information Technology Institute (ITI)
- Machine Learning Specialization
- Python Programming – MaharaTech
- IoT Fundamentals (40 Hours) – Egypt Makes Electronics (EME) / ITIDA, January 2024

PROJECTS

Career Radar – AI Resume Analyzer

- Developed an AI-powered resume analyzer using Python, Streamlit, SentenceTransformers, and Google Gemini
- Implemented semantic similarity matching between resumes and job descriptions using text embeddings
- Generated AI-powered ATS match scores, strengths, missing skills, hiring recommendations, and resume improvement suggestions
- Designed an interactive Streamlit interface for real-time resume evaluation

- Technologies: Python, Streamlit, SentenceTransformers, Google Gemini API, NLP

End-to-End ML Pipeline for Handwritten Digit Recognition

May 2026

- Built and deployed a CNN model using PyTorch achieving 99.47% accuracy on the MNIST dataset
- Developed a Flask backend with image preprocessing including cropping, centering, and padding
- Containerized the application using Docker and automated deployment using Jenkins and SonarQube
- Technologies: PyTorch, Flask, Docker, Jenkins, OpenCV

Automated Crop Disease Detection

May 2026

- Developed an object detection model using YOLOv8 for crop disease detection on the PlantDoc dataset
- Built the complete data preprocessing and training pipeline using Google Colab
- Evaluated model performance through inference testing and validation
- Technologies: Python, YOLOv8, OpenCV, Google Colab

E-commerce Web Application

May 2026

- Developed a scalable backend using Flask following RESTful architecture
- Designed PostgreSQL databases for products, users, and orders
- Implemented secure authentication and backend business logic
- Technologies: Flask, PostgreSQL, SQL

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, JavaScript

Artificial Intelligence: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing (NLP), PyTorch, TensorFlow, YOLOv8, OpenCV, SentenceTransformers, LLMs, Prompt Engineering

Backend Development: Flask, REST APIs

DevOps: Docker, Jenkins, SonarQube, Linux, Ubuntu Server

Databases: PostgreSQL, SQL

Tools: Git, GitHub, Arduino, Google Colab